Remote Volunteer Data Management Standard Operating Procedure for Orthoprojection Segmentation Using TagLab

This has been adapted from Mote Marine Laboratory TagLab Analysis Standard Operating Procedure, more information and SOPs can be found at Mote's SOP GitHub Repository (https://github.com/Mote-Coral-Reef-Restoration/MoteSOPs)

PROJECT SET-UP

1. Your orthomosaics/orthoprojections (from here on out referred to as 'orthos') will be located on a Google Drive under **remoteVolunteers** -> **Gulati**, or at this link (https://drive.google.com/drive/folders/1QVI54ygoGjjWCtTQo3cVsZIyo1zEJHjc?usp=sharing).

2. Here you will find folders corresponding to different orthomosaics. The naming convention will be as follows **YYYY_MM_DD_Site-OutplantEvent**.
    a. For example: If Outplant event 8 occurred at Site S on August 19th, 2022 the folder would read as the following**: 2022_08_19_S-8**.

3. Within these folders you will find a .tif file of the same name as the folder ex: 2022_08_19_S-8.tif. Download the file and store it locally on your machine in a similar file structure as in the google drive.
    a. Ex: Choose a local file path somewhere along the lines of "../MoteAnnotations/2022_08_19_S-8" with the ".." portion being either your Documents or Desktop folder, whichever makes the most sense for you.

4. When you are ready to begin segmentation, open TagLab using either the desktop shortcut or the TagLab.py file located in your local GitHub/TagLab folder.

5. To load an image, select **Project → Add New Map**

6. Fill out the relevant fields
    a. **Map Name:** What you want to call that specific image, this should be whatever your .tif file is named (with the exception of the file extension)
    b. **RGB Image**: this is the file path of where you are keeping the .tif locally, either type in the file path to your image or click the symbol to the right ". . . " and navigate via your file browser
    c. **Depth Image:** We are not using Digital Elevation Models (DEMs) so do not worry about this, if we change our protocols we will update you as well
    d. **Acquisition Date:** Fill in the date for when the imagery was captured format: YYYY-MM-DD, it is already in your file name

e. **Pixel size (mm):** How long one pixel is (mm), do not worry about this, we will scale once the project has been loaded.

Note: Map Name, Path to RGB Image, and Acquisition Date are necessary in order to start your project.

7. Click Apply

8. Save the project by navigating to **File** → **Save As** and then save the project within the same folder as your ortho
    a. Projects are saved as a JSON file (.json)
    b. Name the project the same name as your ortho with your initials at the end in lowercase
        i. ex: **2022_08_19_S-8mg.json**

9. Navigate to **File** → **Settings** and select the box labeled **Autosave**. This will periodically save your project as projectName_autosave.json
    a. We are not relying on the autosave to save our work (please save regularly) however, this can potentially save us hours of duplicative efforts if TagLab crashes

10. You should now be able to see your imported image on screen and are ready to begin segmenting your image.

Image Analysis

1. To set the dictionary, navigate to **Project** → **Labels Dictionary Editor. . .** and select **Load**.
    a. Navigate to where you are storing your dictionary, this should be stored locally and can be found on the Google Drive Folder TagLabDictionary (https://drive.google.com/drive/folders/16fvG4EuANA2XkhxPmFUJzwkwPMp-KVnb?usp=drive_link)
    b. Once loaded, select **Replace**

2. Since scale was not set at the inception of the project, before we begin segmenting we must set the scale. Navigate to the **Measure Tool,** ninth on the list in the left hand tool bar (Shortcut: 9)
    a. If you know your mm/pixel measurements you can input that manually
    b. Use your cursor to select two ends of a known distance, we will be using the center dots of two ends of a scale bar
    c. Enter your known distance in mm (for the scale bar that's 500mm),
    d. Select **Set new pixel size**

    e. Measure that known distance, or another known distance using the same tool to double check that your scale was set correctly

Note: We use scale bars with their centers spaced 50cm apart

3. To set your working area navigate to **Project → Set Working Area**
    a. Select the square with a cursor (to the left of the "Set" button)
    b. Drag the square to encompass your working area

Note: We delineate our working area to encompass all outplants within an event

Note: You can enter your working area manually using local coordinates, but this is not recommended.

4. You can then use the **Create Grid too**l to make a grid to place over the working area. This breaks the whole ortho down into more manageable pieces for more thorough segmentation.
    a. In addition, **Active/Deactivate Grid Operations** allows for boxes of the grid to be marked as complete, incomplete, or empty to help track segmentation progress through the whole ortho.

5. Begin segmenting your image
    a. A list of the tools and functions can be found here (https://taglab.isti.cnr.it/docs)
    b. We tend to use the **Positive/negative clicks segmentation tool**, the **4-clicks segmentation tool**, and the **Edit Border tool** the most (in that order)

6. Once the image is segmented, navigate to **File → Export → Export Annotations as Data Table**
    a. This will export your annotations as a Comma Separated Value file (.csv). Name that CSV the same as your ortho with your initials at the end in lowercase
        i. ex: **2022_08_19_S-8mg.csv**
    b. Save that file within the same local folder as your ortho and project

7. Once you are finished your segmentations, upload your project file (.json) and data (.csv) back into the Google Drive Folder labeled "Data". You DO NOT need to upload the ortho you were working from.